Chapter 9. Factor Analysis

Model, estimation (principal components, principal factors, MLE), rotation, factor scores

Contents

1 The Orthogonal Factor Model

The covariance of p observed variables is explained by a few, k, unobserved common factors.

Definition 1: factor model

$$X_i = \mu_i + \sum_{\ell=1}^{k} \lambda_{i\ell} F_\ell + U_i \quad (i = 1, \dots, p), \qquad \mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F} + \mathbf{U},$$

where $\boldsymbol{\Lambda} = (\lambda_{i\ell})$ is the $p \times k$ **loading matrix**, $\mathbf{F} \in \mathbb{R}^k$ the common factors and $\mathbf{U} \in \mathbb{R}^p$ the specific factors, with

$$\mathrm{E}(\mathbf{F}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{F}) = \mathbf{I}_k, \quad \mathrm{E}(\mathbf{U}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{U}) = \boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \dots, \psi_p), \quad \mathrm{Cov}(\mathbf{F}, \mathbf{U}) = \mathbf{0}.$$

Theorem 2: covariance structure

$$\mathrm{E}(\mathbf{X}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}, \qquad \mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \boldsymbol{\Lambda},$$

$$\mathrm{Var}(X_i) = \underbrace{\sum_{\ell=1}^{k} \lambda_{i\ell}^2}_{h_i^2\text{: communality}} + \underbrace{\psi_i}_{\text{specific variance}}, \qquad \mathrm{Cov}(X_i, X_j) = \sum_{\ell} \lambda_{i\ell}\lambda_{j\ell} \ (i \neq j).$$

Proof. $\mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}) = \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F})\boldsymbol{\Lambda}^\top + \mathrm{Cov}(\mathbf{U}) + \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F}, \mathbf{U}) + \mathrm{Cov}(\mathbf{U}, \mathbf{F})\boldsymbol{\Lambda}^\top = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$, and $\mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}, \mathbf{F}) = \boldsymbol{\Lambda}$. ■

Because $\boldsymbol{\Psi}$ is diagonal, covariances between variables come *entirely* from the common factors. The model explains the $p(p+1)/2$ entries of $\boldsymbol{\Sigma}$ with $pk + p$ parameters.

1.1 Non-identifiability and rotation

For orthogonal $\mathbf{T}$ $(k \times k)$, $\boldsymbol{\Lambda}^* = \boldsymbol{\Lambda}\mathbf{T}$ and $\mathbf{F}^* = \mathbf{T}^\top\mathbf{F}$ give $\boldsymbol{\Lambda}^*\mathbf{F}^* = \boldsymbol{\Lambda}\mathbf{F}$, $\mathrm{Cov}\mathbf{F}^* = \mathbf{I}_k$ and $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top$: loadings are determined only up to rotation. If $\hat{\boldsymbol{\Lambda}}$ is an MLE, so is $\hat{\boldsymbol{\Lambda}}\mathbf{R}$ for any orthogonal $\mathbf{R}$. Removing the $k(k-1)/2$ rotational degrees of freedom leaves $pk + p - k(k-1)/2$ parameters and

$$\nu = \frac{p(p+1)}{2} - \left\{pk + p - \frac{k(k-1)}{2}\right\} = \frac{(p-k)^2 - p - k}{2},$$

which must be positive for the model to impose a real restriction.

1.2 Invariance to scale changes

If $Y_i = c_i X_i$, i.e. $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\mathbf{X}$, then $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\boldsymbol{\mu} + \mathrm{diag}(\mathbf{c})\boldsymbol{\Lambda}\mathbf{F} + \mathrm{diag}(\mathbf{c})\mathbf{U}$ is again a factor model with $\boldsymbol{\Lambda}_Y = \mathrm{diag}(\mathbf{c})\boldsymbol{\Lambda}$ and $\boldsymbol{\Psi}_Y = \mathrm{diag}(\mathbf{c})^2\boldsymbol{\Psi}$. For standardised variables ($\mathrm{Var}X_i = 1$),

$$\lambda_{i\ell} = \mathrm{Cov}(X_i, F_\ell) = \mathrm{Corr}(X_i, F_\ell), \qquad \mathrm{Corr}(Y_i, F_\ell) = \lambda_{i\ell},$$

so loadings are read as variable–factor correlations. In practice factor analysis is often run on the correlation matrix $\mathbf{R}$.

2 Estimation

2.1 Principal component method

From $\widehat{\mathrm{Corr}}(\mathbf{X}) = \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} = \sum_{j=1}^{p} l_j \mathbf{u}_j \mathbf{u}_j^\top \approx \sum_{j=1}^{k} l_j \mathbf{u}_j \mathbf{u}_j^\top$ (largest k eigenvalues),

$$\hat{\mathbf{\Lambda}} = [\sqrt{l_1}\mathbf{u}_1, \ldots, \sqrt{l_k}\mathbf{u}_k], \qquad \hat{\mathbf{\Psi}} = \mathrm{diag}(\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top).$$

The diagonal of $\mathbf{R}$ is reproduced exactly, and the sum of squared off-diagonal residuals of $\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top - \hat{\mathbf{\Psi}}$ is at most $l_{k+1}^2 + \cdots + l_p^2$. Factor ℓ explains a proportion l_ℓ/p of the total variance. Since eigenvalues do not change, increasing k leaves earlier loadings unchanged.

Remark 3: number of factors with the principal component method

Heuristics are used when PCA is used to estimate the factor loadings:

1. use a **scree plot** or a **cumulative scree plot** of $l_1 \geq l_2 \geq \cdots \geq l_p$;
2. or use **Kaiser's rule**: retain factors $1, \ldots, k$ satisfying

$$l_k > \bar{l} = \frac{1}{p}\sum_{i=1}^{p} l_i.$$

When correlation PCA is used, $\sum_i l_i = \mathrm{tr}\mathbf{R} = p$, so $\bar{l} = 1$ and the rule keeps factors with $l_k > 1$ (a factor must explain more than one standardised variable's worth of variance).

These are the same rules as for choosing the number of principal components (Ch. 7); the likelihood ratio test of §?? is available only under the normal MLE approach.

2.2 Principal factor method

Starting from initial communality estimates $\hat{h}_i^2$ (e.g. $1 - 1/r^{ii}$, r^{ii} the diagonal of $\mathbf{R}^{-1}$), form the reduced correlation matrix $\mathbf{R} - \hat{\mathbf{\Psi}}$ and iterate

1. $\mathbf{R} - \hat{\mathbf{\Psi}} \approx \hat{\mathbf{\Lambda}}_{(k)}\hat{\mathbf{\Lambda}}_{(k)}^\top$ (top-k eigendecomposition of the reduced matrix);
2. $\hat{\psi}_{(k)i} = 1 - \sum_{\ell=1}^{k} \hat{\lambda}_{(k)i\ell}^2$

until convergence.

2.3 Maximum likelihood

Assume $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})$. Profiling $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, twice the negative log-likelihood is, up to constants and a factor n,

$$\ell(\hat{\boldsymbol{\mu}}, \mathbf{\Lambda}, \mathbf{\Psi}) = \log|\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi}| + \mathrm{tr}\{\mathbf{S}_0(\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})^{-1}\} + c.$$

For identifiability impose that $\mathbf{\Delta} = \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda}$ is diagonal (if $\hat{\mathbf{\Lambda}}$ is an MLE, so is $\hat{\mathbf{\Lambda}}\mathbf{R}$).

Theorem 4: (∗) $\mathbf{\Lambda}$ given $\mathbf{\Psi}$

Given $\mathbf{\Psi} = \mathrm{diag}(\psi_1, \ldots, \psi_p)$, let $\mathbf{A} = \mathbf{\Psi}^{-1/2}\mathbf{S}_0\mathbf{\Psi}^{-1/2} = \sum_{i=1}^{p} d_i \mathbf{e}_i \mathbf{e}_i^\top$ $(d_1 \geq \cdots \geq d_p)$ and $\delta_i = d_i - 1 > 0$ $(i \leq k)$. Then

$$\hat{\mathbf{\Lambda}} = \mathbf{\Psi}^{1/2}[\mathbf{e}_1, \ldots, \mathbf{e}_k]\mathrm{diag}(\delta_1^{1/2}, \ldots, \delta_k^{1/2}).$$

Proof. Let $\boldsymbol{\Lambda}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}$ and $\boldsymbol{\Sigma}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Sigma}\boldsymbol{\Psi}^{-1/2} = \boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} + \mathbf{I}_p$. Then $\ell = \log|\boldsymbol{\Psi}| + \log|\boldsymbol{\Sigma}^*| + \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1})$. If $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top}$ has eigenvalues $\gamma_1 \geq \cdots \geq \gamma_k \geq 0 = \cdots = 0$, $\boldsymbol{\Sigma}^*$ has eigenvalues $1+\gamma_j$ and 1. Von Neumann's lower bound (pair the small eigenvalues of $\boldsymbol{\Sigma}^{*-1}$ with the large ones of $\mathbf{A}$) gives

$$\log|\boldsymbol{\Sigma}^*| + \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1}) \geq \sum_{j=1}^{k}\Big\{\log(1+\gamma_j) + \frac{d_j}{1+\gamma_j}\Big\} + \sum_{j>k} d_j,$$

with equality when the eigenvectors of $\boldsymbol{\Sigma}^*$ coincide with the $\mathbf{e}_j$ in the same order. Each term $\log x + d/x = \log d + \{d/x - \log(d/x)\} \geq \log d + 1$ (Theorem 3) is minimised at $x = d$, so $1+\gamma_j = d_j$, i.e. $\gamma_j = \delta_j$. Hence $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} = \sum_{j\leq k}\delta_j\mathbf{e}_j\mathbf{e}_j^\top$ and $\boldsymbol{\Lambda}^* = [\mathbf{e}_1,\ldots,\mathbf{e}_k]\operatorname{diag}(\delta_j^{1/2})$ up to rotation; this choice makes $\boldsymbol{\Lambda}^{*\top}\boldsymbol{\Lambda}^* = \boldsymbol{\Delta}$ diagonal. ■

Theorem 5: (**) $\boldsymbol{\Psi}$ given $\boldsymbol{\Lambda}$

The MLE satisfies $\operatorname{diag}(\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}) = \operatorname{diag}(\mathbf{S}_0)$ (the model reproduces the sample variances). Accordingly, given $\boldsymbol{\Lambda}$, update

$$\hat{\boldsymbol{\Psi}} = \operatorname{diag}(\hat{\psi}_1,\ldots,\hat{\psi}_p), \qquad \hat{\psi}_i = (\mathbf{S}_0 - \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)_{ii}.$$

Alternate the two steps. If some $\hat{\psi}_i \leq 0$ (a Heywood case), bound it below by a small positive number.

Logic for (*) and (**): the likelihood equations

We evaluate $\dfrac{\partial\ell(\boldsymbol{\Lambda},\boldsymbol{\Psi})}{\partial\boldsymbol{\Psi}} = 0$ and $\dfrac{\partial\ell(\boldsymbol{\Lambda},\boldsymbol{\Psi})}{\partial\boldsymbol{\Lambda}} = 0$ for

$$\ell(\boldsymbol{\Lambda},\boldsymbol{\Psi}) = \log|\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}| + \operatorname{tr}\{\mathbf{S}_0(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1}\}, \qquad \boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi},$$

using the facts that for a matrix $\mathbf{A} = \mathbf{A}(x)$,

$$\frac{\partial\mathbf{A}^{-1}}{\partial x} = -\mathbf{A}^{-1}\frac{\partial\mathbf{A}}{\partial x}\mathbf{A}^{-1}, \qquad \frac{\partial\log|\mathbf{A}|}{\partial x} = \operatorname{tr}\Big(\mathbf{A}^{-1}\frac{\partial\mathbf{A}}{\partial x}\Big)$$

(the second is the scalar form of Theorem 5; the first follows from differentiating $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$).

Derivative in ψ_i. With $\mathbf{e}_i$ the ith unit vector, $\frac{\partial}{\partial\psi_i}\operatorname{diag}(\psi_1,\ldots,\psi_p) = \mathbf{e}_i\mathbf{e}_i^\top$, so $\frac{\partial\boldsymbol{\Sigma}}{\partial\psi_i} = \mathbf{e}_i\mathbf{e}_i^\top$ and

$$\frac{\partial\ell}{\partial\psi_i} = \underbrace{\operatorname{tr}\Big(\boldsymbol{\Sigma}^{-1}\frac{\partial\boldsymbol{\Sigma}}{\partial\psi_i}\Big)}_{\partial\log|\boldsymbol{\Sigma}|/\partial\psi_i} + \underbrace{\operatorname{tr}\Big(\mathbf{S}_0\frac{\partial\boldsymbol{\Sigma}^{-1}}{\partial\psi_i}\Big)}_{\partial\operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1})/\partial\psi_i} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{e}_i^\top) - \operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1})$$
$$= \mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{e}_i - \mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{e}_i = \mathbf{e}_i^\top(\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1})\mathbf{e}_i = 0, \qquad i = 1,\ldots,p.$$

$$\textcircled{1} \qquad \operatorname{diag}(\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}) = \mathbf{0}.$$

Derivative in λ_{ij}. With $\boldsymbol{\Lambda} = [\mathbf{l}_1,\ldots,\mathbf{l}_k]$ (columns), $\dfrac{\partial\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top}{\partial\lambda_{ij}} = \mathbf{l}_j\mathbf{e}_i^\top + \mathbf{e}_i\mathbf{l}_j^\top$. Then

$$\frac{\partial\ell}{\partial\lambda_{ij}} = \operatorname{tr}\Big(\boldsymbol{\Sigma}^{-1}\frac{\partial\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top}{\partial\lambda_{ij}}\Big) - \operatorname{tr}\Big(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\frac{\partial\boldsymbol{\Sigma}}{\partial\lambda_{ij}}\boldsymbol{\Sigma}^{-1}\Big)$$
$$= \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{l}_j\mathbf{e}_i^\top) + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{l}_j^\top) - \operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{l}_j\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}) - \operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{l}_j^\top\boldsymbol{\Sigma}^{-1})$$
$$= 2\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{l}_j - 2\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{l}_j = 0 \qquad (i = 1,\ldots,p,\ j = 1,\ldots,k),$$

using symmetry of $\boldsymbol{\Sigma}^{-1}$ and $\boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}$ (each pair of traces is equal). Collecting over i, j:

$$\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} \overset{\text{premultiply } \boldsymbol{\Sigma}}{\Longrightarrow} \boldsymbol{\Lambda} = \mathbf{S}_0\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} = \mathbf{S}_0(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top+\boldsymbol{\Psi})^{-1}\boldsymbol{\Lambda} = \mathbf{S}_0\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I}+\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda})^{-1},$$

where the last step uses $\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}) = (\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$, i.e. $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1}\boldsymbol{\Lambda} = \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda})^{-1}$.

Write $\boldsymbol{\Delta} \equiv \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$ (and require $\boldsymbol{\Delta}_{k\times k}$ to be diagonal). Then

$$②' \qquad \boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Delta}) = \mathbf{S}_0\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}.$$

Premultiplying by $\boldsymbol{\Psi}^{-1/2}$:

$$② \qquad \underbrace{\boldsymbol{\Psi}^{-1/2}\mathbf{S}_0\boldsymbol{\Psi}^{-1/2}}_{=:\mathbf{A}}\ \underbrace{\boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}}_{=:\tilde{\mathbf{E}}} = \underbrace{\boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}}_{\tilde{\mathbf{E}}}\ \underbrace{(\mathbf{I} + \boldsymbol{\Delta})}_{=:\mathbf{D}=\mathrm{diag}(d_1,\ldots,d_k)}, \qquad \tilde{\mathbf{E}} = [\tilde{\mathbf{e}}_1, \ldots, \tilde{\mathbf{e}}_k].$$

Hence $\mathbf{A}\tilde{\mathbf{e}}_i = \tilde{\mathbf{e}}_i d_i$, $i = 1, \ldots, k$: $(\tilde{\mathbf{e}}_i, d_i)$ is an eigenpair of $\mathbf{A}$, although $\tilde{\mathbf{e}}_i^\top\tilde{\mathbf{e}}_i \neq 1$ in general. Since $\tilde{\mathbf{e}}_i^\top\tilde{\mathbf{e}}_i = (\tilde{\mathbf{E}}^\top\tilde{\mathbf{E}})_{ii} = (\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda})_{ii} = \Delta_{ii}$, let $\mathbf{e}_i \equiv \tilde{\mathbf{e}}_i/\Delta_{ii}^{1/2}$ so that $\mathbf{e}_i^\top\mathbf{e}_i = 1$ and, for $\mathbf{E} = [\mathbf{e}_1, \ldots, \mathbf{e}_k]$, $\mathbf{E}^\top\mathbf{E} = \mathbf{I}_k$ ($\boldsymbol{\Delta}$ diagonal). Then $(\mathbf{e}_i, d_i)$ is a unit eigenvector–eigenvalue pair of $\mathbf{A}$, and

$$\mathbf{D} = \mathbf{I} + \boldsymbol{\Delta}, \qquad \tilde{\mathbf{E}} = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda} = \mathbf{E}\boldsymbol{\Delta}^{1/2}.$$

$(*)$ **$\boldsymbol{\Psi}$ is fixed.** From ②, $\mathbf{A} = \boldsymbol{\Psi}^{-1/2}\mathbf{S}_0\boldsymbol{\Psi}^{-1/2}$ is computable, with eigenpairs $(\mathbf{e}_i(\mathbf{A}), d_i(\mathbf{A}))$. Use $\mathbf{D} = \mathbf{I} + \boldsymbol{\Delta}$ to set $\hat{\boldsymbol{\Delta}} = \mathrm{diag}(d_i(\mathbf{A})) - \mathbf{I}$, and $\mathbf{E}\boldsymbol{\Delta}^{1/2} = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}$ to set

$$\hat{\boldsymbol{\Lambda}} = \boldsymbol{\Psi}^{1/2}\mathbf{E}(\mathbf{A})\hat{\boldsymbol{\Delta}}^{1/2}, \qquad \mathbf{E}(\mathbf{A}) = [\mathbf{e}_1(\mathbf{A}), \ldots, \mathbf{e}_k(\mathbf{A})],$$

taking the k largest eigenvalues (the von Neumann argument above shows these give the minimum).

$(**)$ **$\boldsymbol{\Lambda}$ is fixed.** From ①, $\mathrm{diag}(\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}) = \mathbf{0}$.

Claim: if ②′ holds, then

$$\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Psi}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1} = \boldsymbol{\Psi}^{-1}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top - \mathbf{S}_0 + \boldsymbol{\Psi})\boldsymbol{\Psi}^{-1}.$$

Proof of claim. $\boldsymbol{\Sigma}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda} = (\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda} = \boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Delta})$, and by ②′ also $\mathbf{S}_0\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda} = \boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Delta})$, so $(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda} = \mathbf{0}$. By Woodbury, $\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Psi}^{-1} - \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}$, hence

$$(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Sigma}^{-1} = (\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1} - \underbrace{(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}}_{=\mathbf{0}}(\mathbf{I} + \boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1} = (\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1},$$

and transposing, $\boldsymbol{\Sigma}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0) = \boldsymbol{\Psi}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)$. Therefore $\boldsymbol{\Sigma}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1} = \boldsymbol{\Psi}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1}$.

Since $\boldsymbol{\Psi}^{-1}$ is diagonal, the ith diagonal entry of $\boldsymbol{\Psi}^{-1}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top - \mathbf{S}_0 + \boldsymbol{\Psi})\boldsymbol{\Psi}^{-1}$ is $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top - \mathbf{S}_0 + \boldsymbol{\Psi})_{ii}/\psi_i^2$, so ① becomes

$$\mathrm{diag}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top - \mathbf{S}_0 + \boldsymbol{\Psi}) = \mathbf{0} \quad \Longrightarrow \quad \psi_i = (\mathbf{S}_0 - \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)_{ii},$$

which is the update $(**)$ and also shows $\mathrm{diag}(\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}) = \mathrm{diag}(\mathbf{S}_0)$ at the MLE.

2.4 Likelihood ratio test for the number of factors

$H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$ (k factors) versus H_1: unrestricted $\boldsymbol{\Sigma}$. With $L_1 = L(\bar{\mathbf{x}}, \mathbf{S}_0)$ and $L_0 = L(\bar{\mathbf{x}}, \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}})$,

$$W = -2\log\frac{L_0}{L_1} = n\log\Big(\frac{|\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}|}{|\mathbf{S}_0|}\Big) \ \dot{\sim}\ \chi^2_\nu,$$

$$\nu = \frac{p(p+1)}{2} - \Big\{pk + p - \frac{k(k-1)}{2}\Big\} = \frac{(p-k)^2 - p - k}{2}.$$

(At the MLE $\mathrm{tr}\{\mathbf{S}_0\hat{\boldsymbol{\Sigma}}^{-1}\} = p$, so the trace terms cancel.) With Bartlett's correction, reject the k-factor model if $(n - 1 - \frac{2p+4k+5}{6})\ln\frac{|\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top+\hat{\boldsymbol{\Psi}}|}{|\mathbf{S}_0|} > \chi^2_\nu(\alpha)$.

3 Factor Rotation

Non-identifiability is used to find interpretable loadings, ideally a simple structure in which each variable loads heavily on only one factor. Rotate $\boldsymbol{\Lambda}^* = \boldsymbol{\Lambda}\mathbf{J}$ with $\mathbf{J}$ orthogonal.

Definition 6: varimax and quartimax

Let $\tilde{\lambda}^*_{i\ell} = \lambda^*_{i\ell}/h_i$ (normalised by the communality).

$$\text{Varimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{\ell=1}^{k} \widehat{\mathrm{Var}}\big((\tilde{\lambda}^*_{1\ell})^2, \ldots, (\tilde{\lambda}^*_{p\ell})^2\big)$$

— spread the squared loadings within each *column* (factor), so each factor has some large and some near-zero loadings.

$$\text{Quartimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{i=1}^{p} \widehat{\mathrm{Var}}\big((\lambda^*_{i1})^2, \ldots, (\lambda^*_{ik})^2\big)$$

— spread the squared loadings within each *row* (variable), so each variable concentrates on one factor.

Rotation leaves the communalities h_i^2 and $\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top$ unchanged.

4 Factor Score Prediction

4.1 Weighted least squares (Bartlett) scores

(1) Choose $\hat{\mathbf{F}}$ to minimise the distance between $\mathbf{X}$ and $\boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F}$: with the estimates fixed, regard $\mathbf{x}_i - \bar{\mathbf{x}} = \hat{\boldsymbol{\Lambda}}\mathbf{F}_i + \mathbf{U}_i$ as a heteroscedastic regression,

$$\hat{\mathbf{F}}_i^{\text{wls}} = \arg\min_{\mathbf{F}_i} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\boldsymbol{\Lambda}}\mathbf{F}_i)^\top \hat{\boldsymbol{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\boldsymbol{\Lambda}}\mathbf{F}_i) = (\hat{\boldsymbol{\Lambda}}^\top \hat{\boldsymbol{\Psi}}^{-1} \hat{\boldsymbol{\Lambda}})^{-1} \hat{\boldsymbol{\Lambda}}^\top \hat{\boldsymbol{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$$

(set the gradient of the quadratic form to zero).

4.2 Regression scores

(2) Since $\begin{pmatrix} \mathbf{X} \\ \mathbf{F} \end{pmatrix} \sim N_{p+k}\Big(\begin{pmatrix} \boldsymbol{\mu} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi} & \boldsymbol{\Lambda} \\ \boldsymbol{\Lambda}^\top & \mathbf{I}_k \end{pmatrix} \Big)$, the conditional expectation (Theorem 3) gives

$$\hat{\mathbf{F}}_i^{\text{reg}} = \mathrm{E}(\mathbf{F} \mid \mathbf{X} = \mathbf{x}_i) = \hat{\boldsymbol{\Lambda}}^\top (\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}})^{-1} (\mathbf{x}_i - \bar{\mathbf{x}}).$$

Theorem 7: relation between the two scores

With $\boldsymbol{\Delta} = \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1} \boldsymbol{\Lambda}$, $\hat{\mathbf{F}}^{\text{reg}} = (\mathbf{I} + \boldsymbol{\Delta})^{-1} \boldsymbol{\Delta}\, \hat{\mathbf{F}}^{\text{wls}} = (\mathbf{I} + \boldsymbol{\Delta}^{-1})^{-1} \hat{\mathbf{F}}^{\text{wls}}$. If $\boldsymbol{\Delta}$ is diagonal each component is shrunk by $\delta_\ell/(1+\delta_\ell) < 1$, so $|\hat{F}_\ell^{\text{wls}}| \geq |\hat{F}_\ell^{\text{reg}}|$.

Proof. By Woodbury, $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1} = \boldsymbol{\Psi}^{-1} - \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1}$, so

$$\boldsymbol{\Lambda}^\top (\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1} = \{\mathbf{I} - \boldsymbol{\Delta}(\mathbf{I} + \boldsymbol{\Delta})^{-1}\} \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1} = (\mathbf{I} + \boldsymbol{\Delta})^{-1} \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1}$$

(as $\mathbf{I} - \boldsymbol{\Delta}(\mathbf{I} + \boldsymbol{\Delta})^{-1} = (\mathbf{I} + \boldsymbol{\Delta} - \boldsymbol{\Delta})(\mathbf{I} + \boldsymbol{\Delta})^{-1}$), while $\boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1}(\mathbf{x} - \bar{\mathbf{x}}) = \boldsymbol{\Delta}\hat{\mathbf{F}}^{\text{wls}}$. ■

Regression scores shrink towards 0, reflecting the prior $\mathbf{F} \sim N(\mathbf{0}, \mathbf{I})$ (a Bayes estimate); WLS scores treat $\mathbf{F}$ as a fixed parameter and are conditionally unbiased.

5 PCA versus Factor Analysis

	PCA	Factor analysis
Goal	linear combinations explaining total variance tr$\mathbf{\Sigma}$	latent factors explaining covariances (off-diagonal)
Model	none (descriptive transformation)	$\mathbf{\Sigma} = \mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi}$
Direction	variables $\rightarrow$ components ($\mathbf{Z} = \mathbf{U}^\top\mathbf{X}$)	factors $\rightarrow$ variables ($\mathbf{X} = \boldsymbol{\mu} + \mathbf{\Lambda}\mathbf{F} + \mathbf{U}$)
Uniqueness	unique if eigenvalues distinct	only up to rotation
Increasing k	earlier components unchanged	(MLE) earlier loadings change